SCSE2103 Data Structures and Algorithms — Lab Exercise 1 Report

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Executive Summary & Task Completion Overview

This report documents the successful implementation, debugging, and verification of the 10 core tasks covering Chapters 1 through 4 of Data Structures and Algorithms (C++). All source code has been thoroughly tested, compiled cleanly, and organized strictly within the designated student directory (./A25CS5003/Lab1/) in compliance with the course submission guidelines.

Task-by-Task Implementation Summary

* Task 1 (Pointer to Object Debugging): Corrected syntax to use the arrow (->) member access operator (ptr->printStudent()).
* Task 2 (GPA Validation Modifier): Implemented setGPA() to validate that newGpa is inclusively between 0.0f and 4.0f before updating.
* Task 3 & 4 (Array Roster & Average GPA): Fixed loop boundaries in printAllStudents and implemented summation logic in calculateAverageGPA.
* Task 5 & 6 (Binary File Operations): Fixed ios::binary flags and write() casting, and implemented random-access reading using seekg().
* Task 7 & 8 (Recursion & Search): Added base cases to recursiveFactorial and implemented recursive linear search in recursiveSearchID.
* Task 9 & 10 (Algorithm Complexity): Fixed nested loop termination in analyzeAlgorithmComplexity and implemented halving loop steps in analyzeLogarithmicEfficiency.